

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 10%

Code of Conduct:

I G.Dhanusha Sree(192325012) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:G.Dhanusha Sree

Rubrics for Evaluation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks(100)
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	

1. An image appears too bright.

Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.

Introduction

An image is said to be **too bright** when most of its pixel intensity values are concentrated toward the higher end of the grey-level range. As a result, important details may be lost in bright regions. A suitable grey-level transformation can reduce the intensity of pixels and improve visibility.

Suitable Technique: Intensity Level Reduction / Logarithmic Transformation

For a very bright image, **intensity scaling** or a suitable grey-level transformation that compresses high intensities can be used. A logarithmic transformation is particularly useful for compressing a wide range of intensity values, although direct intensity reduction is often the simplest practical solution.

The logarithmic transformation is:

$$\begin{bmatrix} s = c \log(1+r) \end{bmatrix}$$

where:

- (r) = input pixel intensity
- (s) = output pixel intensity
- (c) = scaling constant

The transformation maps the input intensity into a new range.

Application Process

1. Read the bright image.
2. Convert the image to grayscale if necessary.
3. Identify the intensity range of the image.
4. Apply the selected grey-level transformation to each pixel.
5. Normalize the resulting values to the required range, such as 0–255.
6. Display the enhanced image.

7. Compare the original and enhanced images.

For a simple intensity reduction, the transformation can be represented as:

$$[s = \alpha r]$$

where ($0 < \alpha < 1$).

For example, if ($\alpha = 0.7$), a pixel with intensity 200 becomes:

$$[s = 0.7(200) = 140]$$

Thus, excessive brightness is reduced.

Expected Outcome

- Overall brightness is reduced.
- Details hidden in excessively bright regions become more visible.
- Pixel intensities are brought into a more useful range.
- The image becomes easier to interpret.

Justification

Intensity reduction is suitable because the main problem is **excessive high intensity**. By mapping high input intensities to lower output values, the dynamic range becomes more suitable for visual interpretation.

Conclusion

Therefore, an appropriate grey-level transformation, particularly **intensity scaling with ($0 < \alpha < 1$)**, can effectively correct an overly bright image and improve the visibility of important details.

2. A dark image lacks detail.

Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.

Introduction

A dark image contains most of its pixel intensities in the **low-intensity region**. Important objects and details may therefore be difficult to distinguish. A transformation that expands low grey levels is required.

Suitable Technique: Power-Law (Gamma) Transformation

The most suitable transformation is **power-law or gamma transformation**:

$$\begin{bmatrix} s = cr^{\gamma} \end{bmatrix}$$

where:

- (r) = input intensity
- (s) = output intensity
- (c) = constant
- (γ) = gamma value

For:

$$\begin{bmatrix} 0 < \gamma < 1 \end{bmatrix}$$

the transformation expands dark intensity values and makes the image brighter.

Working Principle

When ($\gamma < 1$), low input intensity values are mapped to higher output values. Therefore, dark regions become more visible.

For example, assume normalized intensity ($r=0.25$) and ($\gamma=0.5$):

$$\begin{bmatrix} s = (0.25)^{0.5} = 0.5 \end{bmatrix}$$

The intensity increases from 0.25 to 0.5.

Application Process

1. Read the dark image.

2. Convert it to grayscale if required.
3. Normalize pixel values between 0 and 1.
4. Select a gamma value less than 1.
5. Apply:

```
[
s=cr^\gamma
]
```

1. Convert the output back to the required intensity range.
2. Display and compare the enhanced image.

Expected Outcome

- Dark regions become brighter.
- Hidden details become visible.
- Object-background separation improves.
- Overall visibility increases.
- Low-intensity information is enhanced.

Justification

Gamma transformation is suitable because the problem is concentrated in the **low-intensity region**. Unlike simple brightness addition, gamma correction provides a controlled nonlinear transformation and is particularly effective for revealing details in dark areas.

Example

For an 8-bit image, if a pixel has intensity 40, gamma correction with ($\gamma < 1$) maps it to a considerably higher intensity. Thus, objects that were previously difficult to see become clearer.

Conclusion

Therefore, **power-law transformation with gamma less than 1** is an effective method for enhancing dark images because it expands low-intensity regions and improves the visibility of hidden details.

3. An image has poor contrast.

Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.

Introduction

Contrast refers to the difference between the darkest and brightest regions of an image. A low-contrast image contains pixel values concentrated within a narrow range, making objects and boundaries difficult to distinguish.

Suitable Technique: Histogram Equalization

Histogram Equalization is a widely used histogram-based technique for improving global contrast.

A histogram represents the frequency of occurrence of each grey level in an image.

If the histogram is concentrated in a small intensity range, histogram equalization redistributes the intensity values over a wider range.

Basic Principle

Histogram equalization uses the **cumulative distribution function (CDF)** of the image histogram to calculate new intensity values.

A simplified transformation is:

$$\begin{bmatrix} s = (L-1) \text{CDF}(r) \end{bmatrix}$$

where:

- (r) = input intensity
- (s) = output intensity
- (L) = number of grey levels

Application Process

1. Read the image.
2. Convert it into grayscale.
3. Calculate the histogram.
4. Calculate the probability of each intensity.
5. Calculate the cumulative distribution function.
6. Map the original intensities to new intensity values.
7. Generate the enhanced image.
8. Compare the original and enhanced histograms.

Example

Suppose most pixels in an image have intensities between 80 and 150. The image will appear dull because only a small portion of the 0–255 range is used.

After histogram equalization, these intensities are redistributed over a wider range, such as approximately 0–255.

Expected Outcome

- Contrast increases.
- Dark areas become distinguishable.
- Bright areas become clearer.
- Object boundaries become more visible.
- The available grey-level range is used more effectively.

Justification

Histogram equalization is appropriate because poor contrast is directly related to an unsuitable distribution of intensity values. The method automatically redistributes the grey levels based on the image histogram.

Limitation

Global histogram equalization may sometimes over-enhance noise or produce unnatural results. For images with different local lighting conditions, **CLAHE (Contrast Limited Adaptive Histogram Equalization)** can be preferred.

Conclusion

Thus, histogram equalization is a suitable technique for enhancing a low-contrast image because it redistributes pixel intensities and improves the overall visual separation between different regions.

4.A low-contrast image requires uniform intensity distribution. Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.

Introduction

A low-contrast image does not use the available grey-level range effectively. Its intensity values may be concentrated within a narrow range. To obtain a more uniform distribution of intensities, **histogram equalization** can be applied.

Method: Histogram Equalization

Histogram equalization is a contrast enhancement technique that transforms the intensity values so that the resulting histogram is approximately spread across the available intensity range.

Concept

The method uses the probability distribution and cumulative distribution function of image intensities.

The transformation is:

$$\left[\begin{array}{l} s = (L-1) \sum_{j=0}^r p_r(j) \end{array} \right]$$

where:

- (r) = input grey level
- (s) = output grey level
- (L) = total number of grey levels
- $(p_r(j))$ = probability of occurrence of grey level (j)

Working Process

1. Acquire the low-contrast image.
2. Convert it to grayscale.
3. Calculate the histogram.
4. Normalize the histogram.
5. Calculate the cumulative distribution function.
6. Use the CDF to calculate new grey levels.

7. Replace old pixel intensities with transformed values.
8. Obtain the enhanced image.

Effect on Intensity Levels

Before equalization, the intensity values may be concentrated around a narrow range such as 90–140.

After equalization, the values are redistributed across a much wider range.

Before:

[
90,100,110,120,130,140
]

After enhancement:

[
0,45,90,140,200,255
]

The exact values depend on the original histogram.

Expected Result

- More effective use of the available grey-level range.
- Improved contrast.
- Better visibility of objects and boundaries.
- More distinguishable dark and bright regions.
- Improved image quality.

Justification

Histogram equalization is appropriate when the objective is to redistribute intensity values over a wider range. It automatically uses the image histogram to determine the intensity mapping.

Conclusion

Therefore, **histogram equalization** is an appropriate method for a low-contrast image requiring a more uniform use of intensity levels. It improves contrast by redistributing the grey levels across the available range.

5. An image shows uneven brightness across regions.

Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.

Introduction

Uneven brightness occurs when different parts of an image receive different amounts of illumination. For example, one side may be bright while another side is dark. This is known as **non-uniform illumination**.

Suitable Technique: Homomorphic Filtering

Homomorphic filtering is suitable for correcting non-uniform illumination while enhancing image details.

An image can be represented approximately as:

$$\begin{bmatrix} f(x,y) = i(x,y)r(x,y) \end{bmatrix}$$

where:

- $i(x,y)$ = illumination component
- $r(x,y)$ = reflectance component
- $f(x,y)$ = observed image

Illumination generally changes slowly, while reflectance contains details and edges.

Working Principle

Homomorphic filtering separates the illumination and reflectance components in the frequency domain.

Taking logarithm:

$$\begin{bmatrix} \ln f(x,y) = \ln i(x,y) + \ln r(x,y) \end{bmatrix}$$

The image is then transformed using Fourier Transform.

Application Process

1. Acquire the image with uneven illumination.
2. Convert it to grayscale if necessary.
3. Take the logarithm of the image.
4. Apply Fourier Transform.
5. Design a high-pass or homomorphic filter.
6. Suppress low-frequency illumination components.
7. Preserve or enhance high-frequency reflectance components.
8. Apply inverse Fourier Transform.
9. Convert the result back to the normal intensity range.

Expected Outcome

- Uneven background illumination is reduced.
- Dark and bright regions become more balanced.
- Important edges and details are enhanced.
- Overall contrast becomes more uniform.

Example

Consider a document photograph where the top-left region is bright and the bottom-right region is dark. Homomorphic filtering reduces the slowly varying illumination component and makes the document more uniformly illuminated.

Justification

Low-frequency components are mainly associated with gradual illumination changes, whereas high-frequency components contain edges and details. Homomorphic filtering reduces the influence of illumination while preserving important image information.

Conclusion

Thus, **homomorphic filtering** is an effective technique for correcting non-uniform illumination because it separates illumination from reflectance and suppresses unwanted low-frequency brightness variations.

6.A noisy image contains random intensity variations. Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

Introduction

Noise is an unwanted random variation in pixel intensity. It can occur because of sensor limitations, transmission errors, poor lighting, or electronic interference. Smoothing filters are used to reduce such noise.

Suitable Technique: Gaussian Filter

For random noise where smooth noise reduction is required, a **Gaussian filter** is a suitable smoothing filter.

The Gaussian filter uses a weighted average of neighboring pixels.

The two-dimensional Gaussian function is:

$$\begin{bmatrix} G(x,y)=\frac{1}{2\pi\sigma^2} \\ e^{-\frac{x^2+y^2}{2\sigma^2}} \end{bmatrix}$$

where (σ) controls the amount of smoothing.

Working Mechanism

The filter uses a Gaussian kernel. Pixels near the center receive larger weights, while pixels farther away receive smaller weights.

For example, a simplified kernel can be:

$$\begin{bmatrix} \frac{1}{16} \\ \begin{bmatrix} 1&2&1 \\ 2&4&2 \\ 1&2&1 \end{bmatrix} \end{bmatrix}$$

The kernel is moved across the image, and each output pixel is calculated using the weighted average.

Application Process

1. Read the noisy image.
2. Select an appropriate Gaussian kernel size.
3. Choose a suitable (σ).
4. Convolve the kernel with the image.
5. Calculate the weighted average for each pixel.
6. Generate the smoothed image.
7. Compare the noise level before and after filtering.

Expected Outcome

- Random intensity variations are reduced.
- The image becomes smoother.
- Small fluctuations are suppressed.
- Some fine edges may also become slightly blurred.

Justification

Gaussian filtering is effective because it gives greater importance to nearby pixels and gradually decreases the contribution of distant pixels. It produces smooth noise reduction without the abrupt effects sometimes produced by simple averaging.

Conclusion

Therefore, a **Gaussian smoothing filter** is suitable for reducing random noise because it averages neighboring pixels using Gaussian weights and produces a smooth, visually natural result.

7. An image requires noise reduction without significant loss of details. Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.

Introduction

In many images, noise must be removed while important edges and fine structures must be preserved. A simple mean or Gaussian filter can reduce noise but may blur edges.

Suitable Technique: Median Filtering

A **median filter** is particularly effective for removing impulse or salt-and-pepper noise while preserving edges better than linear averaging filters.

Working Principle

The median filter replaces the value of a pixel with the median of the intensity values in its neighborhood.

Consider a 3×3 neighborhood:

```
[  
\begin{bmatrix}  
20&21&22\  
21&255&23\  
20&22&24  
\end{bmatrix}  
]
```

The value 255 is an extreme noisy pixel.

After sorting:

```
[  
20,20,21,21,22,22,23,24,255  
]
```

The median is:

```
[  
22  
]
```

Therefore, the noisy value 255 is replaced by 22.

Application Process

1. Read the noisy image.
2. Select a window, such as 3×3 .
3. Move the window over every pixel.
4. Sort the intensity values within the window.
5. Select the median value.
6. Replace the center pixel with the median.

7. Repeat for the entire image.

Why It Preserves Details

Unlike averaging, the median filter does not calculate an average that can spread extreme values across neighboring pixels. Therefore, sharp transitions such as object boundaries are generally preserved better.

Expected Outcome

- Salt-and-pepper noise is significantly reduced.
- Edges remain relatively sharp.
- Important image structures are preserved.
- The resulting image is cleaner.

Justification

Median filtering is suitable when noise reduction and detail preservation are both important, especially for impulse noise. It removes isolated extreme values without excessively averaging neighboring edge pixels.

Limitation

A very large median-filter window can remove fine details. Therefore, a small window such as 3×3 should normally be selected first.

Conclusion

Thus, the **median filter** provides a good balance between noise removal and edge preservation, making it an appropriate spatial filtering technique for noisy images where details must be retained.

8. An image appears blurred after smoothing.

Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.

Introduction

Smoothing filters reduce noise but may also remove high-frequency components such as edges and fine details. This can make an image appear blurred. A sharpening technique is therefore required to enhance these details.

Suitable Technique: Laplacian Sharpening

The **Laplacian filter** is a second-order derivative operator used for detecting rapid intensity changes and sharpening images.

For a 2-D image:

$$\left[\begin{aligned} \nabla^2 f = & \\ & \frac{\partial^2 f}{\partial x^2} \\ & + \\ & \frac{\partial^2 f}{\partial y^2} \end{aligned} \right]$$

A commonly used Laplacian kernel is:

$$\left[\begin{aligned} & \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} \end{aligned} \right]$$

Working Principle

Edges and fine details correspond to rapid changes in intensity. The Laplacian emphasizes these high-frequency components.

The sharpened image can be obtained using:

$$\begin{bmatrix} g(x,y)=f(x,y)-\nabla^2 f(x,y) \end{bmatrix}$$

depending on the selected kernel convention.

Application Process

1. Take the blurred image.
2. Select a Laplacian kernel.
3. Convolve the kernel with the image.
4. Obtain the Laplacian response.
5. Combine the response with the original blurred image.
6. Generate the sharpened image.
7. Compare the edges and fine details.

Expected Outcome

- Edges become sharper.
- Fine structures become more visible.
- Object boundaries become clearer.
- Blurred appearance is reduced.

Justification

Blurring mainly suppresses high-frequency components. The Laplacian emphasizes rapid intensity changes, allowing lost edge information to be highlighted.

Example

If the boundary between an object and background is weak after smoothing, Laplacian sharpening increases the intensity change around the boundary, making the edge more distinct.

Limitation

Sharpening can also amplify noise. Therefore, the amount of sharpening should be controlled.

Conclusion

Therefore, **Laplacian sharpening** is an effective method for restoring apparent sharpness after smoothing because it emphasizes high-frequency components, particularly edges and fine details.

9. Fine details in an image are not clearly visible.

Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.

Introduction

Fine details such as edges, thin lines, textures, and small structures are mainly represented by **high-frequency components** of an image. When these components are weak, the image appears soft or lacks detail.

Suitable Technique: High-Pass Filtering

A **high-pass filter** is suitable for enhancing fine details because it allows high-frequency components to pass while reducing low-frequency components.

Low-frequency components generally represent:

- Smooth regions
- Gradual intensity changes
- General illumination

High-frequency components represent:

- Edges
- Fine details
- Texture
- Rapid intensity changes

Working Principle

A high-pass filter can be created using a kernel such as:

```
[  
\begin{bmatrix}  
-1&-1&-1\  
-1&8&-1\  
-1&-1&-1  
\end{bmatrix}  
]
```

The filter emphasizes differences between a pixel and its neighboring pixels.

Application Process

1. Read the image.
2. Convert it to grayscale if necessary.
3. Select a high-pass filter.
4. Convolve the filter with the image.
5. Extract the high-frequency components.
6. Add the high-frequency information to the original image if required.
7. Display the enhanced image.

A common sharpening operation is:

$$\begin{bmatrix} g(x,y)=f(x,y)+k h(x,y) \end{bmatrix}$$

where:

- $(f(x,y))$ = original image
- $(h(x,y))$ = high-frequency component
- (k) = sharpening strength
- $(g(x,y))$ = enhanced image

Effect on High-Frequency Components

The high-pass filter emphasizes rapid changes in intensity. Therefore, edges and fine structures become more prominent.

Expected Outcome

- Fine details become clearer.
- Edges become more distinct.
- Texture information becomes more visible.
- Image sharpness increases.

Justification

Since fine details are primarily associated with high-frequency components, enhancing those components directly addresses the problem. High-pass filtering is therefore more appropriate than simple smoothing.

Limitation

Excessive high-pass filtering can amplify noise and create unnatural edges. Therefore, the filter strength should be carefully selected.

Conclusion

Thus, **high-pass filtering** is an appropriate technique for enhancing fine details because it emphasizes high-frequency components corresponding to edges, textures, and small image structures.

10A noisy image must be preprocessed before further analysis. Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

Introduction

Noise can interfere with subsequent image-processing tasks such as segmentation, edge detection, object detection, and classification. Therefore, noise reduction is an important preprocessing step.

Suitable Technique: Median Spatial Filtering

A **median filter** is a suitable spatial filtering technique, particularly when the image contains salt-and-pepper or impulse noise.

Concept

Spatial filtering operates directly on image pixels. A neighborhood is selected around each pixel, and a mathematical operation is performed on the neighborhood to determine the output pixel.

In median filtering, the output is the median value of the neighborhood.

Working Principle

For a 3×3 window:

```
[  
\begin{bmatrix}  
10&11&12\  
11&250&13\  
12&13&14
```

```
\end{bmatrix}
```

Here, 250 is likely a noisy pixel.

After arranging the values:

```
[
10,11,11,12,12,13,13,14,250
]
```

The median is:

```
[
12
]
```

Therefore, the noisy center pixel 250 is replaced with 12.

Application Process

1. Acquire the noisy image.
2. Select a small spatial window such as 3×3 .
3. Place the window over the image.
4. Collect all pixel values inside the window.
5. Sort the values.
6. Find the median.
7. Replace the center pixel with the median.
8. Move the window to the next pixel.
9. Repeat until the complete image is processed.
10. Use the filtered image for further analysis.

Why It Is Suitable

Median filtering is useful because:

- It effectively removes impulse noise.
- It does not spread extreme noise values.
- It preserves edges better than simple averaging.
- It improves the quality of the image before further processing.

Impact on Further Analysis

Noise reduction can improve:

- Edge detection
- Image segmentation
- Feature extraction
- Object recognition
- Classification accuracy

For example, if edge detection is applied directly to a noisy image, random noise may be incorrectly detected as edges. Median filtering reduces such unwanted variations before edge detection.

Expected Outcome

The final image should have:

- Reduced random/impulse noise.
- Better visual quality.
- Preserved important boundaries.
- Improved suitability for subsequent image-processing operations.

Justification

Preprocessing should remove unwanted noise without destroying useful image information. A median spatial filter satisfies this requirement effectively for impulse noise because it replaces abnormal intensity values using neighborhood information while preserving important boundaries.

Conclusion

Therefore, **median spatial filtering** is an effective preprocessing technique for noisy images. It reduces noise, preserves important details, and produces a cleaner image suitable for further analysis.